WORKSHOP SISTEM INFORMASI BERBASIS WEB

MINGGU KE 2

oleh :

MOH.HABIBURROHMAN

E41230411

GOLONGAN : A

PROGRAM STUDI TEKNIK INFORMATIKA JEMBER

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

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Praktikum

Install minikube

```
PS C:\> winget install Kubernetes.minikube
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.minikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-installer.exe
  ██████████████████████████████  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\>
```

Jalankan Minikube

```
PS C:\> minikube start --driver=docker
* minikube v1.38.1 di Microsoft Windows 11 Home Single Language 25H2
* Menggunakan driver docker berdasarkan profil yang ada.
* Memulai node "minikube" dengan peran primary control-plane dalam klaster "minikube"
* Mengunduh image dasar v0.0.50 ...
* Download Kubernetes v1.35.1 preload...
    > index.docker.io/kicbase/sta...:  519.58 MiB / 519.58 MiB  100.00% 917.57
    > preloaded-images-k8s-v18-v1...:  84.73 MiB / 272.45 MiB  31.10% 4.62 KiB ! minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downl
oaded docker.io/kicbase/stable:v0.0.50@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93 as a fallback image
    > preloaded-images-k8s-v18-v1...:  85.20 MiB / 272.45 MiB  31.27% 47 B p/s
PS C:\> minikube start --driver=docker --download-only=false
* minikube v1.38.1 di Microsoft Windows 11 Home Single Language 25H2
* Menggunakan driver docker berdasarkan profil yang ada.
* Memulai node "minikube" dengan peran primary control-plane dalam klaster "minikube"
* Mengunduh image dasar v0.0.50 ...
* Download Kubernetes v1.35.1 preload...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 14.06 M
! minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downloaded docker.io/kicbase/stable:v0.0.50@sha256:eb4fec00e8ad70adf8e6436f195cc4298
25ffb85f95afcdb5d8d9deb576f3e93 as a fallback image
* docker "minikube" container hilang, akan dibuat ulang
* Membuat docker container (CPUs=2, Memory=4000MB) ...
* Menyiapkan Kubernetes v1.35.1 di Docker 29.2.1 ...
* Mengonfigurasi bridge CNI (Container Networking Interface (CNI)) ...
* Memverifikasi komponen Kubernetes...
  - Menggunakan image gcr.io/k8s-minikube/storage-provisioner:v5.
* Aktifkan addons: storage-provisioner, default-storageclass
* Selesai! kubectl sudah dikonfigurasi menggunakan "minikube" klaster dan "default" namespace secara defaul
PS C:\>
```

Cek Node

```
PS C:\> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   41s   v1.35.1
PS C:\>
```

Buat file deployment.yml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Service.yml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

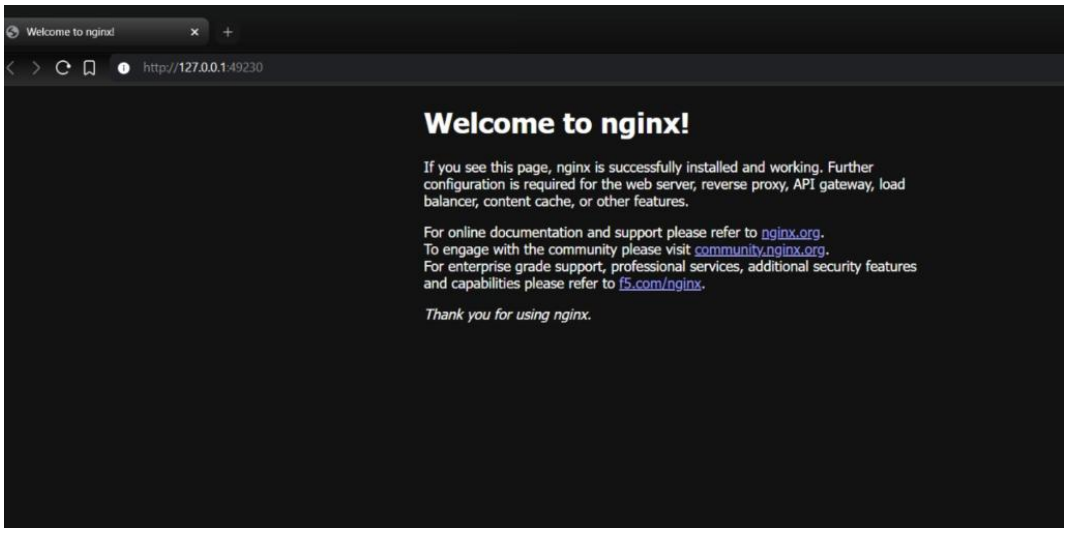
Apply File

```
PS C:\WINDOWS\system32> cd C:\kubernetes
PS C:\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
PS C:\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
PS C:\kubernetes> kubectl get all
NAME                                READY   STATUS              RESTARTS   AGE
pod/web-nginx-6798694bb6-68tkt      0/1     ContainerCreating   0          9s
pod/web-nginx-6798694bb6-cs7d6      0/1     ContainerCreating   0          9s
pod/web-nginx-6798694bb6-vht6w      0/1     ContainerCreating   0          9s

NAME                        TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes          ClusterIP   10.96.0.1      <none>        443/TCP        3m1s
service/web-nginx-service   NodePort    10.105.1.209   <none>        80:31345/TCP   8s

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx   0/3     3            0           9s

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-6798694bb6   3         3         0       9s
PS C:\kubernetes>
```

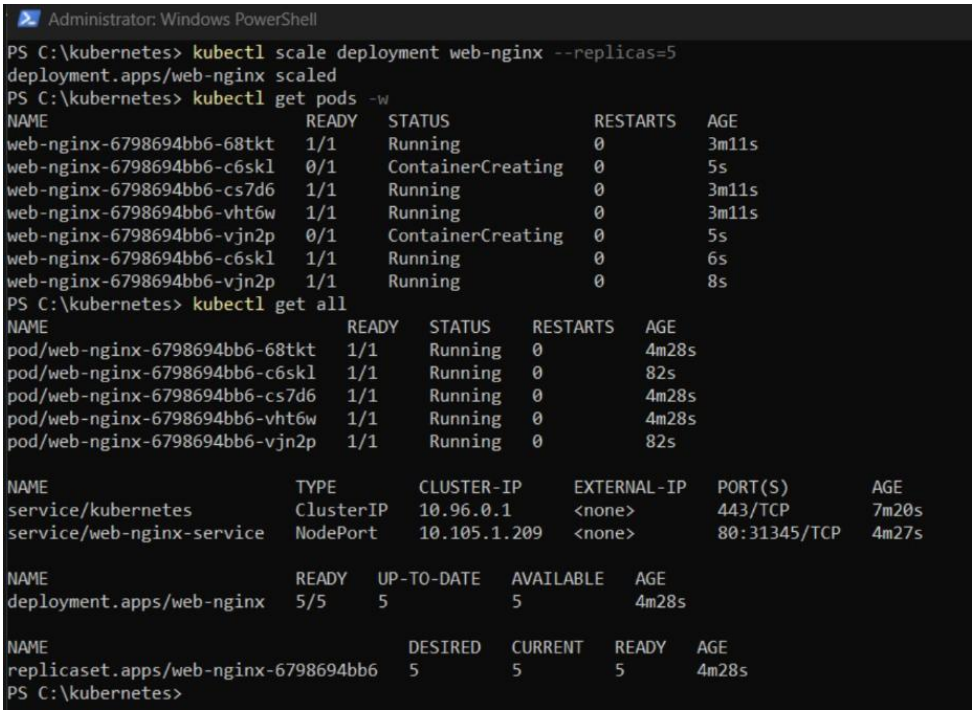
Lihat Semua Resource dan pod lalu jalankan nginx

```
NAME                        TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes          ClusterIP   10.96.0.1      <none>        443/TCP        3m1s
service/web-nginx-service   NodePort    10.105.1.209   <none>        80:31345/TCP   8s

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx   0/3     3            0           9s

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-6798694bb6   3         3         0       9s
PS C:\kubernetes> kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-68tkt          1/1     Running   0          41s
web-nginx-6798694bb6-cs7d6          1/1     Running   0          41s
web-nginx-6798694bb6-vht6w          1/1     Running   0          41s
PS C:\kubernetes> minikube service web-nginx-service
|  NAMESPACE  |        NAME        | TARGET PORT |          URL          |
|-------------|--------------------|-------------|-----------------------|
|  default    | web-nginx-service  |  80         | http://192.168.49.2:31345 |

* Starting tunnel for service web-nginx-service.

|  NAMESPACE  |        NAME        | TARGET PORT |          URL          |
|-------------|--------------------|-------------|-----------------------|
|  default    | web-nginx-service  |             | http://127.0.0.1:49230 |

* Opening service default/web-nginx-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

Nginx berjalan



Scaling pod Up ke 5



```
Administrator: Windows PowerShell
PS C:\kubernetes> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
PS C:\kubernetes> kubectl get pods -w
NAME                          READY   STATUS              RESTARTS   AGE
web-nginx-6798694bb6-68tkt    1/1     Running             0          3m11s
web-nginx-6798694bb6-c6skl    0/1     ContainerCreating   0          5s
web-nginx-6798694bb6-cs7d6    1/1     Running             0          3m11s
web-nginx-6798694bb6-vht6w    1/1     Running             0          3m11s
web-nginx-6798694bb6-vjn2p    0/1     ContainerCreating   0          5s
web-nginx-6798694bb6-c6skl    1/1     Running             0          6s
web-nginx-6798694bb6-vjn2p    1/1     Running             0          8s
PS C:\kubernetes> kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/web-nginx-6798694bb6-68tkt    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-c6skl    1/1     Running   0          82s
pod/web-nginx-6798694bb6-cs7d6    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-vht6w    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-vjn2p    1/1     Running   0          82s

NAME                       TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP        7m20s
service/web-nginx-service  NodePort    10.105.1.209   <none>        80:31345/TCP   4m27s

NAME                       READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx  5/5     5            5           4m28s

NAME                                  DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-6798694bb6  5         5         5       4m28s
PS C:\kubernetes>
```

Scale Down ke 3

```
Administrator: Windows PowerShell
web-nginx-6798694bb6-cs7d6   1/1     Running             0          3m11s
web-nginx-6798694bb6-vht6w   1/1     Running             0          3m11s
web-nginx-6798694bb6-vjn2p   0/1     ContainerCreating   0          5s
web-nginx-6798694bb6-c6skl   1/1     Running             0          6s
web-nginx-6798694bb6-vjn2p   1/1     Running             0          8s
PS C:\kubernetes> kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/web-nginx-6798694bb6-68tkt    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-c6skl    1/1     Running   0          82s
pod/web-nginx-6798694bb6-cs7d6    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-vht6w    1/1     Running   0          4m28s
pod/web-nginx-6798694bb6-vjn2p    1/1     Running   0          82s

NAME                      TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
service/kubernetes        ClusterIP   10.96.0.1      <none>        443/TCP        7m20s
service/web-nginx-service NodePort    10.105.1.209   <none>        80:31345/TCP   4m27s

NAME                      READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx 5/5     5            5           4m28s

NAME                                 DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-6798694bb6 5         5         5       4m28s
PS C:\kubernetes> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
PS C:\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-68tkt    1/1     Running   0          4m57s
web-nginx-6798694bb6-cs7d6    1/1     Running   0          4m57s
web-nginx-6798694bb6-vht6w    1/1     Running   0          4m57s
```

Menampilkan node cluster.

```
Administrator: Windows PowerShell
PS C:\kubernetes> minikube node add
* Adding node m02 to cluster minikube as [worker]
! Cluster was created without any CNI, adding a node to it might cause broken networking.
* Starting "minikube-m02" worker node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Creating docker container (CPUs=2, Memory=2200MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
* Successfully added m02 to minikube!
PS C:\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE     VERSION
minikube       Ready      control-plane   9m52s   v1.35.1
minikube-m02   NotReady   <none>          18s     v1.35.1
PS C:\kubernetes>
```

Delete

```
Administrator: Windows PowerShell
PS C:\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS C:\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
PS C:\kubernetes> minikube stop   # menghentikan VM/container
* Stopping node "minikube-m02"  ...
* Powering off "minikube-m02" via SSH ...
* Stopping node "minikube"  ...
* Powering off "minikube" via SSH ...
* 2 nodes stopped.
PS C:\kubernetes> minikube delete # menghapus cluster
* Deleting "minikube" in docker ...
* Deleting container "minikube" ...
* Deleting container "minikube-m02" ...
* Removing C:\Users\snNHQ\.minikube\machines\minikube ...
* Removing C:\Users\snNHQ\.minikube\machines\minikube-m02 ...
* Removed all traces of the "minikube" cluster.
PS C:\kubernetes>
```